ECO 310: Precept 3

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Outline

1. Question 1: Compensated Demand
2. Question 2: Income and Substitution Effects
3. Question 3: Expected Utility

Question 1: Statement

Given the utility function

$$u(x_1, x_2) = \sqrt{x_1} + \sqrt{x_2},$$

find the **compensated (Hicksian) demand** and the **expenditure function**.

Key Concepts: What is Compensated Demand?

- The compensated (Hicksian) demand for a good is the bundle that achieves a given utility level u^* **at the lowest possible expenditure**.
- **Why it matters:**
  - It separates out *substitution effects* from *income effects*.
  - It is used to derive the *expenditure function*, $e(p_1, p_2, u^*)$.
- **Approach:** We solve

$$\min_{x_1, x_2} \; p_1 x_1 + p_2 x_2 \quad \text{s.t.} \quad u(x_1, x_2) = u^*.$$

- **Outcome:**

$$x_i^c = x_i^c(p_1, p_2, u^*), \quad e(p_1, p_2, u^*) = p_1 x_1^c + p_2 x_2^c.$$

Solution Steps

1. Equalize Marginal Utility per Dollar:

$$\frac{MU_1}{MU_2} = \frac{p_1}{p_2}, \quad MU_1 = \frac{1}{2\sqrt{x_1}}, \quad MU_2 = \frac{1}{2\sqrt{x_2}}.$$

$$\implies \sqrt{x_2} = \frac{p_1}{p_2}\sqrt{x_1}.$$

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2. Impose Utility Target u^*:

$$\sqrt{x_1} + \sqrt{x_2} = u^*.$$

Substitute $\sqrt{x_2}$:

$$\sqrt{x_1} + \frac{p_1}{p_2}\sqrt{x_1} = u^* \quad \implies \quad \sqrt{x_1} = \frac{p_2}{p_1 + p_2}\, u^*.$$

Solution Steps (Cont.)

3. Solve for $\sqrt{x_2}$ and Square:

$$\sqrt{x_2} = \frac{p_1}{p_1 + p_2} u^*, \quad x_1^c = \left(\frac{p_2}{p_1 + p_2} u^* \right)^2, \quad x_2^c = \left(\frac{p_1}{p_1 + p_2} u^* \right)^2.$$

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4. Expenditure Function:

$$e(p_1, p_2, u^*) = p_1 x_1^c + p_2 x_2^c = \frac{p_1 p_2}{p_1 + p_2} (u^*)^2.$$

Solution Steps (Cont.)

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Answer:

$$\boxed{x_1^c = \left(\frac{p_2}{p_1 + p_2} u^* \right)^2, \quad x_2^c = \left(\frac{p_1}{p_1 + p_2} u^* \right)^2, \quad e(p_1, p_2, u^*) = \frac{p_1 p_2}{p_1 + p_2} (u^*)^2.}$$

Question 2: Statement

Use the same utility function $u(x_1, x_2) = \sqrt{x_1} + \sqrt{x_2}$.

Find the **substitution effect, compensating variation, and income effects** if $p_2 = 1$, and p_1 increases from 1 to 2 while $m = 6$. Explain what this means in words.

You may take as given that the "regular" (Marshallian) demand function is as follows (this is what you'd get by solving (1) $\frac{MU_1}{MU_2} = \frac{p_1}{p_2}$ and (2) $p_1 x_1 + p_2 x_2 = m$):

$$(x_1, x_2)(p_1, p_2, m) = \left(\frac{p_2}{p_1} \frac{m}{p_1 + p_2}, \frac{p_1}{p_2} \frac{m}{p_1 + p_2} \right) \tag{3}$$

Key Concepts: Income vs. Substitution Effects

- **Marshallian Demand** (uncompensated):

$$(x_1, x_2)(p_1, p_2, m) = \left(\frac{p_2}{p_1} \frac{m}{p_1 + p_2}, \ \frac{p_1}{p_2} \frac{m}{p_1 + p_2} \right).$$

- **Substitution Effect:** How x_1 changes if prices change *but* utility is held constant (compensated).
- **Income Effect:** Additional change due to loss (or gain) of purchasing power once prices change *and* income remains the same.
- **Compensating Variation (CV):** The extra money needed to maintain original utility after a price change.

Solution Steps (Income & Substitution Effects)

1. **Original bundle (old prices):** $p_1 = 1, p_2 = 1, m = 6$

$$(x_1^o, x_2^o) = \left(\frac{1}{1} \cdot \frac{6}{1+1}, \frac{1}{1} \cdot \frac{6}{1+1}\right) = (3, 3).$$

$$u^o = \sqrt{3} + \sqrt{3} = 2\sqrt{3}.$$

2. **Final bundle (new prices):** $p_1' = 2, p_2 = 1, m = 6$

$$(x_1^f, x_2^f) = \left(\frac{1}{2}\frac{6}{2+1}, \frac{2}{1}\frac{6}{2+1}\right) = (1, 4).$$

Solution Steps (Cont.)

3. Compensated (Hicksian) bundle:

$$\min_{x_1, x_2} \{p_1' x_1 + p_2 x_2 : u(x_1, x_2) = u^o\}.$$

Using the earlier formula with $p_1' = 2$, $p_2 = 1$, and $u^o = 2\sqrt{3}$,

$$(x_1^c, x_2^c) = \left(\left(\frac{1}{2+1} \cdot 2\sqrt{3} \right)^2, \ \left(\frac{2}{2+1} \cdot 2\sqrt{3} \right)^2 \right) = \left(\frac{4}{3}, \frac{16}{3} \right).$$

4. Decompose x_1 changes:

$$\text{Substitution Effect: } x_1^c - x_1^o = \frac{4}{3} - 3 \approx -1.667.$$

$$\text{Income Effect: } x_1^f - x_1^c = 1 - \frac{4}{3} \approx -0.333.$$

$$\text{Total: } -1.667 + (-0.333) = -2.$$

Solution Steps (Compensating Variation)

- **CV:** Additional income needed at new prices $(p_1' = 2, p_2 = 1)$ to achieve old utility $u^o = 2\sqrt{3}$.
- Expenditure function at new prices:

$$e(2, 1, 2\sqrt{3}) = \frac{2 \cdot 1}{2+1}(2\sqrt{3})^2 = \frac{2}{3} \times 12 = 8.$$

- Original income $m = 6$, so

$$\text{CV} = 8 - 6 = 2.$$

- **Interpretation:** The consumer needs \$2 more to stay at utility $2\sqrt{3}$ after p_1 doubles from 1 to 2.

Question 3: Statement

Sally has preferences over lotteries on $\{0, \$1, \$5\}$. Show that if the independence axiom holds, and if she feels $(0, 1, 0) \succ (.2, 0, .8)$, then she also feels $(0, 1, 0) \succ (.1, .5, .4)$.

Hint: consider mixing each lottery with $(0, 1, 0)$.

Key Concepts: Independence Axiom

- **Independence Axiom (Expected Utility):**
  If $p \succ q$, then for any lottery r and any $\alpha \in (0, 1)$,

$$\alpha p + (1 - \alpha) r \ \succ \ \alpha q + (1 - \alpha) r.$$

- **Interpretation:** A preference between two lotteries does not change if both lotteries are "mixed" with the same third lottery.

Solution Steps (Expected Utility)

- Let
$$p = (0, 1, 0), \quad q = (0.2, 0, 0.8), \quad r = (0, 1, 0).$$
- Given $p \succ q$. By independence, for any $\alpha \in (0, 1)$,
$$\alpha p + (1 - \alpha) r \succ \alpha q + (1 - \alpha) r.$$
- Choose $\alpha = 0.5$. Then
$$0.5\, p + 0.5\, r = (0, 1, 0),$$
$$0.5\, q + 0.5\, r = (0.1, 0.5, 0.4).$$
- Therefore, $(0, 1, 0) \succ (0.1, 0.5, 0.4)$.